

Maharashtra Public Service Commission Assessment

Question Paper Name :	CHROMA1A Applied Mechanics QP2
Subject Name :	Assistant and Associate Professor Applied Mechanics Maharashtra Engineering Teachers Services
Creation Date :	2026-07-22 10:56:16
Duration :	60
Number of Questions :	100
Share Answer Key With Delivery Engine :	No
Actual Answer Key :	Yes

Assistant and Associate Professor Applied Mechanics Maharashtra Engineering Teachers Services

Group Number :	1
Group Id :	4410092970405
Group Maximum Duration :	0
Group Minimum Duration :	60
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0

Assistant and Associate Professor Applied Mechanics Maharashtra Engineering Teachers Services

Section Id :	4410093000104
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100

Number of Questions to be attempted :	100
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	4410093011082
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 4410092111350 Question Type : MCQ Option Shuffling : No

For a beam with a circular cross-section subjected to a transverse shear force, where does the maximum shear stress occur?

Options :

1. At the topmost fiber of the cross-section
2. At the extreme bottom fiber of the cross-section
3. At the neutral axis of the cross-section
4. At a distance of $d/4$ from the neutral axis

Question Number : 2 Question Id : 4410092111625 Question Type : MCQ Option Shuffling : No

Which combination of stresses arises due to eccentric loading on a structural member?

Options :

1. Shear stress and torsional stress
2. Shear stress and bending stress
3. Axial stress and bending stress
4. Axial stress and shear stress

Question Number : 3 Question Id : 4410092112441 Question Type : MCQ Option Shuffling : No

Which of the following statements best describes the function of the shear centre in an unsymmetrical beam cross-section?

Options :

1. It is the point where bending moment becomes zero.
2. It is always located at the centroid for all sections.
3. It is the point where a transverse load causes no twisting of the section.
4. It is the point where the maximum shear stress occurs.

Question Number : 4 Question Id : 4410092111620 Question Type : MCQ Option Shuffling : No

In a prismatic cantilever beam subjected to a uniformly distributed load over its entire span, at which section does the maximum flexural stress occur?

Options :

1. At the free end
2. At mid-span
3. At one quarter span
4. At the fixed support

Question Number : 5 Question Id : 4410091563167 Question Type : MCQ Option Shuffling : No

A simply supported beam carries a bending moment of 2000 N-m at midspan. If its section modulus is 50 cm³, what is the maximum bending stress?

Options :

1. 40 MPa
2. 20 MPa
3. 80 MPa
4. 10 MPa

Question Number : 6 Question Id : 4410091563158 Question Type : MCQ Option Shuffling : No

Which of the following conditions indicates a pure shear state when interpreted from principal strains obtained using a rosette?

Options :

1. Principal strains are equal in magnitude and opposite in sign
2. Principal strains are both positive, but not equal in magnitude
3. Principal strains are both negative, and equal in magnitude
4. Principal strains are zero

Question Number : 7 Question Id : 441009950254 Question Type : MCQ Option Shuffling : No

A thin plate under biaxial stress: $\sigma_x = 80$ MPa (tensile), $\sigma_y = 40$ MPa (compressive). What is σ on plane inclined 0° to x-axis?

Options :

1. 40 MPa tensile
2. 80 MPa tensile
3. 40 MPa compressive
4. 0 MPa

Question Number : 8 Question Id : 4410091527389 Question Type : MCQ Option Shuffling : No

On the plane of maximum shear stress in a two-dimensional stress state, the normal stress is equal to:

Options :

1. Zero
2. Higher principal stress
3. Lower principal stress
4. Average of principal stresses

Question Number : 9 Question Id : 4410092111622 Question Type : MCQ Option Shuffling : No

Which quantity is directly used in the principal stress equations to determine the maximum principal stress at a point in a loaded shaft?

Options :

1. Length of the shaft under applied loading
2. Normal and shear stresses at a point
3. Diameter of the shaft cross-section only
4. Material properties of the shaft section

Question Number : 10 Question Id : 4410092111788 Question Type : MCQ Option Shuffling : No

A ductile shaft is designed using the maximum shear stress theory. If the allowable shear stress is kept unchanged, increasing the shaft diameter slightly will primarily:

Options :

1. Increase the torque-carrying capacity significantly
2. Reduce the modulus of rigidity substantially
3. Increase the working shear stress significantly
4. Eliminate torsional deformation completely

Question Number : 11 Question Id : 4410092112134 Question Type : MCQ Option Shuffling : No

If a shaft is subjected to a torque that varies rapidly with time, which design consideration becomes most critical?

Options :

1. Yield strength under tension force
2. Fatigue strength due to fluctuating torsional stress
3. Deflection caused by steady bending moment
4. Shear stress under vertical shear

Question Number : 12 Question Id : 441009750871 Question Type : MCQ Option Shuffling : No

For a simply supported beam of span 'L' m, carrying a point load 'P' kN at mid-span, the maximum bending moment is:

Options :

1. $PL/4$ kN-m
2. $PL/2$ kN-m
3. $PL/8$ kN-m
4. $PL/6$ kN-m

Question Number : 13 Question Id : 4410091371712 Question Type : MCQ Option Shuffling : No

A simply supported beam of span 5 m carries a UDL of 3 kN/m.
Find the shear force 1 m from the left support.

Options :

1. 3 kN
2. 4.5 kN
3. 6 kN
4. 9 kN

Question Number : 14 Question Id : 4410092111896 Question Type : MCQ Option Shuffling : No

A simply supported beam with a rectangular cross-section of width 100 mm and height 200 mm carries a uniformly distributed load of 10 kN/m over a span of 6 meters. What is the maximum shear stress developed in the cross-section?

Options :

1. 2.25 MPa
2. 0.75 MPa
3. 4.40 MPa
4. 3.50 MPa

Question Number : 15 Question Id : 441009976393 Question Type : MCQ Option Shuffling : No

For a thin-walled cylindrical pressure vessel, which of the following best describes the reason for longitudinal strain development?

Options :

1. Restraint of circumferential deformation
2. Pressure acting on the end caps of the cylinder
3. External pressure on the curved surface
4. Thermal expansion of the cylinder wall

Question Number : 16 Question Id : 4410092112444 Question Type : MCQ Option Shuffling : No

For a given material and the initial diameter, how does the longitudinal strain in a thin cylinder change if the internal pressure is doubled?

Options :

1. Longitudinal strain becomes four times larger
2. Longitudinal strain becomes twice as large
3. Longitudinal strain remains the same
4. Longitudinal strain becomes half as large

Question Number : 17 Question Id : 4410092111353 Question Type : MCQ Option Shuffling : No

A thin-walled cylindrical pressure vessel of internal diameter 400 mm and wall thickness 5 mm is subjected to an internal gauge pressure of 2 MPa. What is the longitudinal (axial) stress in the cylinder wall?

Options :

1. 80 MPa
2. 40 MPa
3. 20 MPa
4. 10 MPa

Question Number : 18 Question Id : 4410092112136 Question Type : MCQ Option Shuffling : No

Which property of a material determines the amount of thermal stress induced in a constrained member when its temperature changes ?

Options :

1. Coefficient of friction
2. Modulus of elasticity
3. Density
4. Poisson's ratio

Question Number : 19 Question Id : 441009950240 Question Type : MCQ Option Shuffling : No

For a uniaxially loaded member, the maximum principal stress is equal to:

Options :

1. Zero
2. Twice the applied stress
3. The applied normal stress
4. The applied shear stress

Question Number : 20 Question Id : 4410092112313 Question Type : MCQ Option Shuffling : No

If the instantaneous cross-sectional area of a tensile specimen becomes 20% less than its original area, the true stress will be:

Options :

1. Less than the engineering stress
2. Greater than the engineering stress
3. Equal to the engineering stress
4. Independent of engineering stress

Question Number : 21 Question Id : 4410092111791 Question Type : MCQ Option Shuffling : No

A compressive load acts eccentrically on a rectangular section. If the eccentricity is gradually increased while the load remains constant, the maximum compressive stress will:

Options :

1. Decrease because the direct stress becomes smaller
2. Increase because the bending stress becomes larger
3. Remain constant because the applied load is unchanged
4. Become zero because the section resists bending fully

Question Number : 22 Question Id : 4410092111466 Question Type : MCQ Option Shuffling : No

A simply supported beam carrying a point load at midspan experiences maximum bending stress at which location?

Options :

1. At the centroid of the beam section at midspan
2. At the extreme fibers of the beam section at midspan
3. At the neutral axis of the beam section at midspan
4. At the supports of the beam

Question Number : 23 Question Id : 4410092112381 Question Type : MCQ Option Shuffling : No

A long steel column carries a compressive load acting slightly away from its centroidal axis. Which of the following structural response in column should the designer expect?

Options :

1. The column experiences combined compression and bending
2. The column behaves as a perfectly short compression member
3. The column develops only uniform axial compression stresses
4. The column shows a significant increase in elastic stiffness

Question Number : 24 Question Id : 4410092111355 Question Type : MCQ Option Shuffling : No

The slenderness ratio of a column is defined as the ratio of its effective length to which of the following geometric properties?

Options :

1. Cross-sectional area
2. Least radius of gyration
3. Moment of inertia about the major axis
4. Depth of the cross-section

Question Number : 25 Question Id : 4410092111011 Question Type : MCQ Option Shuffling : No

For the same axial load, a column with greater eccentricity is generally:

Options :

1. Subjected to lower compressive stresses
2. More resistant to buckling
3. Unaffected by end conditions
4. More susceptible to bending stresses

Question Number : 26 Question Id : 4410092112138 Question Type : MCQ Option Shuffling : No

Which method is used to determine the slope at a point on a loaded beam from the area under the M/EI diagram?

Where, M = Bending Moment at a section of the beam.

E = Modulus of Elasticity (Young's Modulus)

I = Moment of Inertia (Second Moment of Area) of the beam's cross-section.

Options :

1. Double integration method
2. Moment area method
3. Conjugate beam method
4. Graphical method

Question Number : 27 Question Id : 4410092111899 Question Type : MCQ Option Shuffling : No

When applying the conjugate beam method to a cantilever beam of length 4 meters with a point load of 15 kN at the free end, which boundary condition is used for the conjugate beam at the fixed end of real beam?

Options :

1. Free end
2. Roller end
3. Fixed end
4. Pinned end

Question Number : 28 Question Id : 4410092111793 Question Type : MCQ Option Shuffling : No

If a cantilever beam is subjected to a uniformly varying load (UVL) starting from zero at the free end, which statement best describes the deflection shape?

Options :

1. The beam remains straight without any noticeable deflection
2. The beam bends downward with maximum deflection at the free end
3. The beam bends downward with maximum deflection at midspan
4. The beam bends upward with maximum deflection at the fixed end

Question Number : 29 Question Id : 4410092111468 Question Type : MCQ Option Shuffling : No

If the modulus of elasticity of the material is halved for a cantilever beam under UDL, what effect does this have on the maximum deflection at the free end?

Options :

1. The maximum deflection at the free end remains unchanged
2. The maximum deflection at the free end decreases
3. The maximum deflection at the free end doubles
4. The maximum deflection at the free end halves

Question Number : 30 Question Id : 4410092112383 Question Type : MCQ Option Shuffling : No

A simply supported beam carries a single concentrated load. The load is shifted from the midspan to one-quarter of the span without changing its magnitude. What happens to the maximum beam deflection?

Options :

1. It increases after moving the load
2. It decreases after moving the load
3. It remains unchanged after shifting
4. It becomes zero after shifting

Question Number : 31 Question Id : 4410091371794 Question Type : MCQ Option Shuffling : No

A simply supported steel beam of span $L = 4$ m carries a point load $W = 20$ kN at the midspan. If the beam has a moment of inertia $I = 8 \times 10^6$ mm⁴ and $E = 200$ GPa, the maximum deflection is approximately:

Options :

1. 16.67 mm
2. 62.5 mm
3. 1.25 mm
4. 25.0 mm

Question Number : 32 Question Id : 4410092111013 Question Type : MCQ Option Shuffling : No

For a simply supported beam carrying a uniformly distributed load over its entire span, the maximum deflection occurs at the:

Options :

1. Mid-span
2. Left support
3. Quarter-span point
4. Right support

Question Number : 33 Question Id : 4410091371779 Question Type : MCQ Option Shuffling : No

A cantilever of length 2.5 m carries a clockwise moment of 12 kN·m at free end. $EI = 500$ kN·m².

Tip deflection is:

Options :

1. 0.065 m
2. 0.07 m
3. 0.075 m
4. 0.08 m

Question Number : 34 Question Id : 4410092111794 Question Type : MCQ Option Shuffling : No

Which of the following explains why a three-hinged arch is statically determinate under static loading?

Options :

1. It has one hinge at a support and fixed restraint elsewhere
2. It has no hinges and relies only on its curved geometry
3. It has hinges at both supports and one at the crown
4. It has one crown hinge with both supports fully fixed

Question Number : 35 Question Id : 4410092112140 Question Type : MCQ Option Shuffling : No

A loaded three-hinged arch of uniform material is subjected to a uniform temperature increase. What is the primary structural response?

Options :

1. There is vertical displacement at the crown slightly
2. There is a reduction in the span of the arch
3. There is no effect on the arch due to temperature change
4. There is an increase in horizontal thrust at the supports

Question Number : 36 Question Id : 4410092112384 Question Type : MCQ Option Shuffling : No

A flexible cable is supported at both ends and carries a single concentrated load at an intermediate point along its span. Which shape will the cable adopt under this loading?

Options :

1. A horizontal straight line between both supports only
2. Two straight cable segments meeting at the load point
3. Several straight cable segments of equal inclination only
4. A continuous parabolic curve throughout the entire span

Question Number : 37 Question Id : 4410092112449 Question Type : MCQ Option Shuffling : No

If three concentrated vertical loads are applied to a cable at equal distances, what can be said about the horizontal component of tension in all segments between the supports?

Options :

1. The horizontal component of tension is zero at midspan
2. The horizontal component of tension remains constant throughout the cable
3. The horizontal component of tension increases at every load point
4. The horizontal component of tension decreases at every load point

Question Number : 38 Question Id : 4410092111795 Question Type : MCQ Option Shuffling : No

For a suspension bridge cable subject to substantial temperature variations, which design approach would best reduce temperature-induced stresses in the cable system?

Options :

1. Applying suitable prestress to the cable before installation
2. Incorporating expansion joints to permit cable movement
3. Selecting cable material with a higher elastic modulus
4. Increasing the cable cross-sectional area uniformly

Question Number : 39 Question Id : 4410092112021 Question Type : MCQ Option Shuffling : No

If a frame is externally indeterminate to the first degree, what does this imply?

Options :

1. The structure has one more internal force than external force
2. It can be solved by equilibrium equations without extra analysis
3. The number of unknown support reactions exceeds equilibrium equations by two
4. The number of unknown support reactions exceeds equilibrium equations by one

Question Number : 40 Question Id : 4410092112450 Question Type : MCQ Option Shuffling : No

A two-bay, two-story rigid frame is subjected to both vertical and horizontal loads. What is the primary reason this frame is considered statically indeterminate?

Options :

1. The continuity of joints alone causes indeterminacy.
2. The presence of both vertical and horizontal loads causes indeterminacy.
3. The number of unknown reactions is lesser than the available equilibrium equations.
4. The number of unknown reactions and internal forces exceeds available equilibrium equations.

Question Number : 41 Question Id : 4410092111630 Question Type : MCQ Option Shuffling : No

In a rigid-jointed frame, what does kinematic indeterminacy represent?

Options :

1. Total number of external loads applied to the structure
2. Total number of support reactions in the structure
3. Type of engineering material used for the members
4. Total number of independent joint displacements possible

Question Number : 42 Question Id : 4410092111471 Question Type : MCQ Option Shuffling : No

A pin-jointed frame is to be modified to increase its degree of static indeterminacy by one without changing the support conditions. Which modification will achieve this?

Options :

1. Add one extra member between two existing joints
2. Remove one member from a redundant panel
3. Change orientation of a diagonal member only
4. Introduce an internal hinge at a joint location

Question Number : 43 Question Id : 4410092112254 Question Type : MCQ Option Shuffling : No

For a planar truss system to be statically determinate, which criterion must hold true regarding its joints and members?

Options :

1. The number of members must exceed twice the number of joints.
2. The number of members must be equal to the number of joints.
3. The number of members must be equal to the sum of number of joints and support reactions.
4. The sum of the number of members and support reactions must be equal to twice the number of joints.

Question Number : 44 Question Id : 4410092112142 Question Type : MCQ Option Shuffling : No

For a statically determinate simply supported beam subjected to a concentrated transverse load, which combination of equilibrium conditions is sufficient to uniquely evaluate the unknown support reactions?

Options :

1. Equilibrium of horizontal forces and rotational equilibrium about a reference point
2. Equilibrium of vertical forces and rotational equilibrium about a reference point
3. Equilibrium of vertical forces and internal force compatibility relationships
4. Equilibrium of horizontal forces and internal force compatibility relationships

Question Number : 45 Question Id : 4410092111472 Question Type : MCQ Option Shuffling : No

In a frame with more external reactions than equilibrium equations, what must be applied to solve for all reactions?

Options :

1. Considering only the force directions at supports.
2. Examining the type of load distribution applied.
3. Compatibility conditions for deformations at supports.
4. Counting the total number of joints present.

Question Number : 46 Question Id : 4410092111632 Question Type : MCQ Option Shuffling : No

When analyzing a propped cantilever beam subjected to a uniform load, which compatibility condition must be satisfied at the prop location to solve for the unknown reaction?

Options :

1. The slope at the prop must remain zero throughout
2. The vertical deflection at the prop must be zero
3. The bending moment at the prop must remain zero
4. The shear force at the prop must remain zero

Question Number : 47 Question Id : 4410092112255 Question Type : MCQ Option Shuffling : No

According to Castigliano's theorem, which quantity is differentiated with respect to an applied load to determine the corresponding displacement in a statically indeterminate structure?

Options :

1. Flexural rigidity of the members
2. Strain energy of the system
3. Support settlement function
4. Magnitude of reaction forces

Question Number : 48 Question Id : 4410092112143 Question Type : MCQ Option Shuffling : No

When applying the moment distribution method to a continuous beam with varying moment of inertia, which step ensures equilibrium is maintained at each joint?

Options :

1. Successive distribution and carry-over of moments at each joint until moments converge.
2. Summing moments at all supports after the first distribution cycle only.
3. Initial calculation of fixed-end moments for each member before any distribution.
4. Assignment of moment capacities to each joint based on support conditions only.

Question Number : 49 Question Id : 4410092111797 Question Type : MCQ Option Shuffling : No

In a statically indeterminate structure exhibiting internal indeterminacy, compatibility conditions are primarily used to ensure which of the following?

Options :

1. Member forces satisfy equilibrium throughout the entire structural system
2. Connected members undergo compatible deformations at every joint
3. Support reactions remain constant under all possible loading conditions
4. Internal member forces remain equal throughout the entire structure

Question Number : 50 Question Id : 4410092111631 Question Type : MCQ Option Shuffling : No

If a planar frame has four external reactions and only three independent equilibrium equations, what is the degree of its external indeterminacy?

Options :

1. Three
2. One
3. Four
4. Zero

Question Number : 51 Question Id : 4410092112024 Question Type : MCQ Option Shuffling : No

Which of the following factors most negatively impacts the bond strength of concrete?

Options :

1. Reduction in bar diameter
2. Proper compaction of concrete
3. Presence of oil or grease on bars
4. Use of higher grade concrete

Question Number : 52 Question Id : 4410092111018 Question Type : MCQ Option Shuffling : No

The maximum permissible shear stress in concrete specified by IS 456:2000 mainly depends on the:

Options :

1. Effective span of the beam
2. Diameter of longitudinal bars
3. Support conditions of the beam
4. Grade of concrete

Question Number : 53 Question Id : 4410092112144 Question Type : MCQ Option Shuffling : No

Which aspect of bond development between concrete and reinforcing steel is most directly affected by chemical admixture that delay setting time?

Options :

1. Initial development of microstructure at the interface
2. Cross-sectional area of the reinforcing steel
3. Ultimate tensile strength of the steel
4. Mechanical anchorage parameters

Question Number : 54 Question Id : 4410092111633 Question Type : MCQ Option Shuffling : No

In the design of a reinforced concrete beam, which assumption is made regarding the bond between steel and concrete?

Options :

1. Elastic slip occurs at the interface
2. Perfect bond exists between steel and concrete
3. Bond is only partial between steel and concrete
4. No bond exists between steel and concrete

Question Number : 55 Question Id : 4410091372774 Question Type : MCQ Option Shuffling : No

In the limit-state flexural design using IS 456 stress-block, the use of a rectangular (or equivalent) stress block implicitly assumes that _____.

Options :

1. Concrete tensile stresses are resisting bending at ultimate.
2. Steel stress–strain curve is nonlinear and must be iteratively solved.
3. The actual concrete stress is uniformly distributed in the compression zone.
4. The resultant compressive force acts at a known lever arm from the neutral axis.

Question Number : 56 Question Id : 4410092112145 Question Type : MCQ Option Shuffling : No

Which condition best indicates a balanced reinforced concrete beam section based on the depth of the neutral axis?

Options :

1. The depth of neutral axis results in simultaneous yielding of steel and crushing of concrete
2. The depth of neutral axis is solely limited by code provisions without relation to material strengths
3. The depth of neutral axis ensures only steel yields before concrete crushes
4. The depth of neutral axis is always minimized for all beam sections

Question Number : 57 Question Id : 4410091363138 Question Type : MCQ Option Shuffling : No

In a singly reinforced concrete T-beam, the flange contributes to bending resistance only if :

Options :

1. Neutral axis lies below the flange
2. Neutral axis lies entirely within the flange thickness
3. Flange thickness is less than web thickness
4. Flange is not reinforced

Question Number : 58 Question Id : 4410092111800 Question Type : MCQ Option Shuffling : No

A singly reinforced rectangular beam is designed exactly as a balanced section. At ultimate load, which condition exists?

Options :

1. Steel reaches yield strain as concrete reaches ultimate compressive strain
2. Concrete reaches crushing strain before steel reaches yielding strain
3. Steel reaches fracture strain before concrete reaches crushing strain
4. Concrete and steel remain within their respective elastic strain limits

Question Number : 59 Question Id : 4410092112258 Question Type : MCQ Option Shuffling : No

According to IS 456:2000, the minimum tensile reinforcement required for a beam using Fe 415 steel is approximately what percentage of bd (breadth $\times$ effective depth)?

Options :

1. 0.30 percent
2. 0.25 percent
3. 0.15 percent
4. 0.20 percent

Question Number : 60 Question Id : 4410092111020 Question Type : MCQ Option Shuffling : No

For improving the flexural capacity of a reinforced concrete beam without significantly increasing its width, the MOST effective modification is to:

Options :

1. Increase the effective depth
2. Increase the unsupported span of the beam
3. Increase the aggregate size
4. Increase the clear cover

Question Number : 61 Question Id : 4410092112146 Question Type : MCQ Option Shuffling : No

Which of the following best describes the minimum clear cover requirement for a reinforced concrete beam exposed to moderate environmental conditions as per IS 456?

Options :

1. A clear cover of 30 mm to main reinforcement is specified
2. A clear cover of 15 mm to main reinforcement is specified
3. A variable cover based on bar diameter is specified
4. A clear cover of 40 mm to main reinforcement is specified

Question Number : 62 Question Id : 4410092111801 Question Type : MCQ Option Shuffling : No

Which of the following is primarily evaluated under the limit state of serviceability for reinforced concrete beams?

Options :

1. Deflection control
2. Shear failure
3. Ultimate moment capacity
4. Bond strength

Question Number : 63 Question Id : 441009163074 Question Type : MCQ Option Shuffling : No

In case of RC members, the modification factor for tension reinforcement depends on the _____.

Options :

1. grade of concrete
2. grade of steel and the percentage of tension reinforcement
3. shear reinforcement provided
4. span of the beam

Question Number : 64 Question Id : 441009163105 Question Type : MCQ Option Shuffling : No

Which of the following is NOT a valid assumption in the design of reinforced concrete columns subjected to eccentric compression?

Options :

1. Plane sections before bending remain plane after bending.
2. Concrete resists tensile loads.
3. Perfect bond exists between concrete and steel. So, they act together, and no slip can occur between the two materials.
4. Compression load is transferred uniformly.

Question Number : 65 Question Id : 441009143291 Question Type : MCQ Option Shuffling : No

When can a structural frame be safely assumed to be fully braced?

Options :

1. The frame is constructed without any lateral load-resisting elements.
2. Special bracing elements such as shear walls and shear trusses are provided to resist lateral loads.
3. Only vertical loads are considered in design.
4. The frame has flexible connections at all joints.

Question Number : 66 Question Id : 441009642548 Question Type : MCQ Option Shuffling : No

Which of the following assumptions is made in the design of a short reinforced concrete column under axial loading?

Options :

1. Buckling is always considered
2. Concrete is assumed to fail before steel
3. Strain in reinforcement and concrete is the same in axial loading
4. Poisson's ratio is neglected for concrete

Question Number : 67 Question Id : 4410092112260 Question Type : MCQ Option Shuffling : No

What is the principal effect of increasing the slenderness ratio in a column?

Options :

1. It increases the risk of lateral instability and buckling
2. It increases the torsional capacity of the member
3. It makes the column more ductile under load
4. It enhances the compressive strength significantly

Question Number : 68 Question Id : 4410092112392 Question Type : MCQ Option Shuffling : No

A reinforced concrete column has an effective length that is less than twelve times its least lateral dimension. How should this column be classified?

Options :

1. A short reinforced concrete column
2. A long reinforced concrete column
3. A column carrying only axial compression
4. A column requiring only minimum reinforcement

Question Number : 69 Question Id : 4410092111022 Question Type : MCQ Option Shuffling : No

As per IS 456:2000, when the diameter of the main reinforcement in a column exceeds 12 mm, the nominal cover shall be at least:

Options :

1. Equal to the diameter of the bar or 40 mm, whichever is greater
2. Equal to the diameter of the bar or 25 mm, whichever is greater
3. Not more than 40 mm in any case
4. Half the diameter of the bar or 25 mm, whichever is greater

Question Number : 70 Question Id : 4410092112029 Question Type : MCQ Option Shuffling : No

According to IS 456:2000, under which condition is a column with helical reinforcement permitted to have enhanced axial load capacity compared with a tied column?

Options :

1. The volumetric ratio of helical reinforcement satisfies the code requirements
2. The unsupported length of the column exceeds the specified code limit
3. The cross-section of the column is provided in a square shape
4. The concrete used in the column satisfies the minimum grade requirement

Question Number : 71 Question Id : 4410092112261 Question Type : MCQ Option Shuffling : No

As per IS 456:2000, the pitch of transverse reinforcement in a tied reinforced concrete column shall not exceed the least of which of the following?

Options :

1. Least lateral dimension of column, 300 mm, or 16 times the smallest diameter of longitudinal bar
2. Least lateral dimension of column, 250 mm, or 12 times the largest diameter of longitudinal bar
3. Greater lateral dimension of column, 300 mm, or 16 times the smallest diameter of longitudinal bar
4. Least lateral dimension of column, 400 mm, or 20 times the diameter of tie bar

Question Number : 72 Question Id : 4410091373456 Question Type : MCQ Option Shuffling : No

Reinforcement in the waist slab of RCC stairs is primarily provided to _____.

Options :

1. Support handrails
2. Resist bending due to self-weight, superimposed load and imposed load
3. Reduce weight of the stair
4. Control temperature and shrinkage stresses in the inclined slab

Question Number : 73 Question Id : 4410092111638 Question Type : MCQ Option Shuffling : No

While designing a one-way slab as per IS 456:2000, what is the maximum spacing allowed between main reinforcement bars, considering both code provisions and practical requirements for crack control and serviceability?

Options :

1. The lesser of three times the effective depth or 300 mm
2. Two times the effective depth
3. The greater of three times the effective depth or 300 mm
4. 350 mm

Question Number : 74 Question Id : 4410092111911 Question Type : MCQ Option Shuffling : No

Which of the following statements regarding slab design as per IS 456:2000 are correct?

- I. For one-way slabs, the minimum overall depth is generally governed by width-to-depth ratio
- II. For one-way slabs, the minimum overall depth is generally governed by span-to-depth ratio.
- III. The mild steel reinforcement in either longitudinal direction shall not be less than 0.15% of the total cross-sectional area of slab.
- IV. The HYSD steel reinforcement in either longitudinal direction shall not be less than 0.15% of the total cross-sectional area of slab.

Options :

1. II and IV only
2. I and III only
3. II and III only
4. I and IV only

Question Number : 75 Question Id : 4410092112262 Question Type : MCQ Option Shuffling : No

When designing a slab for serviceability, what is the main reason to limit the maximum deflection?

Options :

1. To maximize ultimate strength of the slab
2. To prevent damage to non-structural elements and finishes
3. To maintain yield strength of reinforcement
4. To ensure minimum slab thickness is not exceeded

Question Number : 76 Question Id : 4410092111480 Question Type : MCQ Option Shuffling : No

As per IS 456:2000, which of the following is the primary parameter used for preliminary deflection control in the design of a two-way slab system?

Options :

1. Span-to-depth ratio limitations for the slab
2. Spacing of main reinforcement bars in one direction
3. Stiffness of edge beams along the perimeter
4. Minimum concrete cover requirements for durability

Question Number : 77 Question Id : 4410092112150 Question Type : MCQ Option Shuffling : No

Two simply supported reinforced concrete slabs have the same span, loading, material properties, and the same area of tensile reinforcement (A_{st}), but different overall depths. According to IS 456:2000, which slab is most likely to experience greater deflection and why?

Options :

1. The slab with lesser depth due to reduced flexural stiffness under service loads
2. Both slabs deflect equally due to identical span and reinforcement area
3. The slab with greater depth due to increased cracking under service loads
4. The slabs shall not deflect if designed properly

Question Number : 78 Question Id : 4410091362410 Question Type : MCQ Option Shuffling : No

As per IS 456:2000, in Limit State Design the partial safety factor for concrete (γ_m) considered when assessing the strength of a structure is _____.

Options :

1. 1.15
2. 1.25
3. 1.5
4. 2.0

Question Number : 79 Question Id : 44100925570 Question Type : MCQ Option Shuffling : No

For structural steels without a well-defined yield point, the characteristic strength is taken as the stress at which the permanent set reaches:

Options :

1. 0.1% proof strain
2. 0.2% proof strain
3. 0.3% proof strain
4. 0.4% proof strain

Question Number : 80 Question Id : 4410092112460 Question Type : MCQ Option Shuffling : No

A designer is evaluating two grades of reinforcing steel with similar yield strengths but different ultimate tensile strengths and strain capacities. Which parameter should primarily influence the choice for a ductile design approach in limit state design?

Options :

1. Higher elongation at maximum load before rupture
2. Greater modulus of elasticity for initial stiffness
3. Higher ratio of yield to ultimate strength
4. Higher ultimate tensile strength with lower yield strength

Question Number : 81 Question Id : 4410092111806 Question Type : MCQ Option Shuffling : No

Which assumption permits the superposition of stresses due to prestressing force and external bending moments in the elastic analysis of a prestressed concrete beam?

Options :

1. Concrete tensile strength is equal to compressive strength
2. Prestress force remains constant under all loading conditions
3. Material behavior remains linear within the service load range
4. Tendons and concrete act independently after transfer

Question Number : 82 Question Id : 4410092111913 Question Type : MCQ Option Shuffling : No

Which of the following assumptions are adopted in the elastic analysis of pre-tensioned prestressed concrete members?

1. The cross-section remains plane before and after bending.
2. The tensile strength of concrete is fully considered in flexure.
3. Losses in prestress are neglected.
4. The bond between steel and concrete is perfect.

Options :

1. 1 and 4 only
2. 2 and 3 only
3. 1, 3 and 4 only
4. 2, 3 and 4 only

Question Number : 83 Question Id : 4410092112032 Question Type : MCQ Option Shuffling : No

In the load balancing method for prestressed concrete, the shape of the tendon profile in a simply supported beam is chosen primarily to:

Options :

1. Minimize the overall weight of the structure
2. Match the moment diagram due to external loading
3. Ensure equal stress distribution across depth
4. Maximize the prestressing force at the supports

Question Number : 84 Question Id : 4410092111026 Question Type : MCQ Option Shuffling : No

When the pressure line lies within the kern zone of a prestressed concrete section, which of the following is true?

Options :

1. No tensile stress develops in the concrete section
2. Maximum tensile stress develops at the extreme fibre
3. Shear stresses become zero
4. Prestress losses are eliminated

Question Number : 85 Question Id : 4410092111481 Question Type : MCQ Option Shuffling : No

When analyzing the losses in prestress, which combination of factors should be considered to ensure accurate prediction of the effective prestress in a beam?

Options :

1. Elastic shortening, creep, shrinkage, and relaxation of steel must all be accounted for
2. Anchor set and temperature changes are the exclusive causes of prestress loss
3. Only creep and shrinkage are significant for effective prestress loss
4. Elastic shortening and relaxation are the only factors affecting prestress

Question Number : 86 Question Id : 4410092112264 Question Type : MCQ Option Shuffling : No

A prestressed concrete beam has a rectangular cross-section 120 mm wide and 300 mm deep. The beam is concentrically prestressed by a cable carrying a force of 180 kN. The direct compressive stress produced by the prestressing force is:

Options :

1. 5 N/mm^2
2. 15 N/mm^2
3. 10 N/mm^2
4. 2 N/mm^2

Question Number : 87 Question Id : 4410092112033 Question Type : MCQ Option Shuffling : No

Why is high strength concrete necessary to minimize prestress losses in beams?

Options :

1. It reduces creep and shrinkage effects
2. It requires higher cement content
3. It improves color of finished surface
4. It increases air content in concrete

Question Number : 88 Question Id : 44100925645 Question Type : MCQ Option Shuffling : No

What is the key advantage of using high-strength steel in prestressed concrete beams?

Options :

1. Increased ductility
2. Reduced reinforcement congestion
3. Higher thermal expansion
4. Reduced resistance to corrosion

Question Number : 89 Question Id : 441009146669 Question Type : MCQ Option Shuffling : No

Which of the following statements about pre-tensioning systems is correct?

Options :

1. The tendons are tensioned after the casting of concrete in the moulds.
2. Special anchorages are always required to transfer pre-stress in pre-tensioned members.
3. Tendons are not bonded to concrete after releasing the jacking pressure.
4. High early strength concrete is used to allow quick reuse of moulds.

Question Number : 90 Question Id : 4410092111642 Question Type : MCQ Option Shuffling : No

In a pre-tensioned concrete beam, what is the main purpose of releasing the tension in tendons after concrete hardening?

Options :

1. To increase the tensile strength of the tendons
2. To induce compressive force in the concrete
3. To allow the concrete to expand and contract freely
4. To reduce the modulus of elasticity of the concrete

Question Number : 91 Question Id : 441009672588 Question Type : MCQ Option Shuffling : No

For steel beams (as per IS 800: 2007), the design bending strength of a laterally supported compact section is primarily governed by its _____.

Options :

1. Shear area of web
2. Plastic section modulus (Z_p)
3. Radius of gyration
4. Slenderness ratio

Question Number : 92 Question Id : 4410092112463 Question Type : MCQ Option Shuffling : No

In a slip-critical bolted connection using High Strength Friction Grip bolts, what is the main purpose of applying a specified minimum pretension to the bolt?

Options :

1. To reduce bolt relaxation over time
2. To increase the yield strength of the bolt
3. To ensure friction development between the bolt contact surfaces
4. To improve the ductility of the bolted joint

Question Number : 93 Question Id : 4410092111484 Question Type : MCQ Option Shuffling : No

An engineer is tasked with designing a welded joint in a steel truss where fatigue is a concern due to repeated loading cycles. What measure should be adopted to enhance the fatigue strength of the welded connection?

Options :

1. Changing the orientation of the welds only
2. Using thicker steel plates at the connection
3. Increasing the weld leg size beyond required minimum
4. Ensuring smooth transition and proper weld profile at joints

Question Number : 94 Question Id : 4410092111809 Question Type : MCQ Option Shuffling : No

For a long double-angle strut designed according to IS 800:2007, the governing design consideration is generally:

Options :

1. Yield strength of steel in pure compression
2. Bearing strength of the gusset plate
3. Buckling strength based on member slenderness
4. Shear strength of the angle legs

Question Number : 95 Question Id : 4410092112155 Question Type : MCQ Option Shuffling : No

According to IS 800:2007, while determining the design strength of a battened built-up column, which of the following design step is considered?

Options :

1. Decreased slenderness due to the battening system
2. Increased slenderness due to the battening system
3. Decreased cross sectional area due to the battening system
4. Increased tensile stress resistance due to the battening system

Question Number : 96 Question Id : 441009270901 Question Type : MCQ Option Shuffling : No

According to IS 800:2007, the effective length of a prismatic column effectively held in position and restrained against rotation at both ends is _____.

Options :

1. 0.65 times the unsupported length
2. 0.75 times the unsupported length
3. 1.0 times the unsupported length
4. greater than 1.0 times the unsupported length

Question Number : 97 Question Id : 4410092111485 Question Type : MCQ Option Shuffling : No

In a bolted tension member connection, how does the gusset plate contribute to the joint's performance?

Options :

1. It eliminates the need for bolts
2. It reduces the length of the tension member
3. It increases the strength of bolts used
4. It distributes stresses over a larger area

Question Number : 98 Question Id : 4410092111810 Question Type : MCQ Option Shuffling : No

Why does IS 800:2007 require verification of rupture strength based on the net sectional area of a tension member?

Options :

1. Gross section yielding always occurs after rupture of steel
2. Net area controls the buckling resistance of the member
3. Bolt holes reduce the effective area available to resist tensile force
4. Block shear failure cannot occur in connected members

Question Number : 99 Question Id : 44100953705 Question Type : MCQ Option Shuffling : No

Which of the following is a typical example of a variable action in structural design?

Options :

1. Dead load of the structure.
2. Weight of fixed equipment.
3. Live load on a floor.
4. Self-weight of structural members.

Question Number : 100 Question Id : 441009674594 Question Type : MCQ Option Shuffling : No

What is the primary objective of the Limit State Design method?

Options :

1. To minimize cost of construction
2. To ensure structure fails just before collapse
3. To ensure safety and serviceability under all loading conditions
4. To allow maximum deformation of structure